

Lecture 2: Wealth and Capital Taxation in Ramsey Model: Chamley-Judd

Abdoulaye Ndiaye

NYU

December 24, 2025

Application of Diamond-Mirrlees to Intertemporal Taxation

- ▶ C. Chamley, "Optimal Taxation of Capital Income in General Equilibrium with Infinite Lives", *Econometrica*, 54 (1986), 607-622. reached by;
- ▶ Representative infinitely lived agent with utility $\sum_{t=0}^{\infty} \beta^t u(c_t, l_t)$
- ▶ Simplifying assumptions:
 - ▶ single-agent, infinitely lived
 - ▶ no uncertainty
- ▶ Government:
 - ▶ needs to finance g_t
 - ▶ chooses
 - ▶ taxes to finance g_t
 - ▶ government debt b_t

Equilibrium (I)

Consumer problem (1)

$$\max_{c, l \geq 0, k} \sum_{t=0}^{\infty} \beta^t u(c_t, l_t)$$

s.t.

$$(1 + \tau_{ct})c_t + k_{t+1} + b_{t+1} \leq (1 - \tau_{kt})(1 + r_t - \delta)k_t + (1 - \tau_{lt})w_t l_t + R_t b_t$$

$$k_0 = \bar{k}_0$$

$$l_t \leq 1$$

Equilibrium (II)

► **Firm's problem (2)**

$$\max_{k,l} F(k_t, l_t) - w_t l_t - r_t k_t$$

where F is CRS.

► **Government budget constraint (3)**

$$g_t + R_t b_t \leq \tau_{lt} w_t l_t + \tau_{kt} (1 + r_t - \delta) k_t + \tau_{ct} c_t + b_{t+1}$$

► **Market clearing (4)**

$$c_t + g_t + k_{t+1} \leq F(k_t, l_t) + (1 - \delta) k_1$$

Equilibrium (III)

Definition: CE with taxes $\{\tau_{lt}, \tau_{kt}, \tau_{ct}\}$ and government purchases $\{g_t\}$ is allocations $\{c_t, l_t, k_t, b_t\}$ and prices $\{w_t, r_t\}$ s.t.

1. consumers take prices $\{w_t, r_t\}$ as given and solve consumer problem (1)
 2. firms take prices $\{w_t, r_t\}$ as given , solve producer's problem (2) and make 0 profit in equilibrium.
 3. government's budget constraint is satisfied (3)
 4. markets clear (4)
- ▶ Government needs to find $\pi = \{\tau_{lt}, \tau_{kt}, \tau_{ct}\}$ to finance g_t and maximize $\sum_{t=0}^{\infty} \beta^t u(c_t(\pi), l_t(\pi))$ where $c_t(\pi), l_t(\pi)$ are equilibrium allocations with taxes.
 - ▶ Some observations to what will follow.

Observation 1

- Irrelevance of some taxes: FOCs

$$\frac{u_l(t)}{u_c(t)} = - \frac{(1 - \tau_{lt})w_t}{1 + \tau_{ct}}$$

$$\frac{\beta u_c(t+1)}{u_c(t)} = \frac{p_{t+1}}{p_t} = \frac{(1 + \tau_{ct+1})}{(1 + \tau_{ct})(1 - \tau_{kt+1})(1 + r_{t+1} - \delta)}$$

$$(1 - \tau_{kt+1})(1 + r_{t+1} - \delta) = R_{t+1}$$

- Too many taxes, can get rid of some.
- We will assume no consumption tax, $\tau_{ct} = 0$ for all t
- Equivalently, we could assume that $\tau_{kt} = 0$ and have

$$\frac{(1 + \tilde{\tau}_{ct+1})}{(1 + \tilde{\tau}_{ct})} = (1 - \tau_{kt+1})$$

- Positive tax on capital and constant tax on consumption is equivalent to zero tax on capital and increasing tax on consumption.

- ▶ **Observation 2:** Nothing fancy about dynamics Instead of thinking about period t consumption, think about period 0 consumption of a good with label " t ": equivalent to the static commodity taxation problem with infinitely many goods.
- ▶ **Observation 3:** Non-distortionary taxation of capital in period 0
 - ▶ Note that tax on capital in period 0 does not distort any decisions: equivalent to a lump sum tax.
 - ▶ If government could use this tax, it would set it at a very high level to get enough revenues to finance all future g_t
 - ▶ Assume (without any justification) that this tax is unavailable to make the problem interesting $\tau_{k0} = 0$
- ▶ **Observation 4:** (as before) Many ways to ensure that distortions hold
 - ▶ Here: tax gross return on capital $1 + r - \delta$
 - ▶ Could instead (as usually done in practice) tax net return $r - \delta$: nothing changes in the analysis.

Primal Approach

- ▶ Adding up consumer and government budget constraints gives

$$g_t + c_t + k_{t+1} \leq w_t l_t + (1 + r_t - \delta)k_t = F(k_t, l_t) + (1 - \delta)k_t$$

which is just the resource constraint.

- ▶ Now we can just apply the primal approach by finding the implementability constraint:
- ▶ **Exercise:** Show that the implementability condition is:

$$\sum_{t=0}^{\infty} \beta^t [u_c(t)c_t + u_l(t)l_t] = u_c(0)R_0k_0 \quad [\text{ImC}]$$

- ▶ **Hint:** Solving b_t forward in the budget constraint of the agent to obtain a Period 0 budget constraint then substitute FOCs into budget constraint.
- ▶ so the problem of the planner is

$$\max \sum \beta^t u(c_t, l_t)$$

s.t. the resource constraint and the implementability constraint, ImC.

Zero Capital Taxation in Steady State

- FOCs for $t > 0$

$$\beta^t u_c(t) + \beta^t \mu [u_{cc}(t)c_t + u_c(t) - u_{cl}(t)l_t] = \lambda_t$$

$$\lambda_t = [F_k(t+1) + (1 - \delta)]\lambda_{t+1}$$

Therefore

$$\frac{u_c(t) + \mu [u_{cc}(t)c_t + u_c(t) - u_{cl}(t)l_t]}{u_c(t+1) + \mu [u_{cc}(t+1)c_{t+1} + u_c(t+1) - u_{cl}(t+1)l_{t+1}]} = \beta(F_k(t+1) + (1 - \delta))$$

- **Theorem: No taxes in steady state**

- **Proof.** Suppose $c_t \rightarrow c, k_t \rightarrow k, l_t \rightarrow l$. Then equation above says

$$\beta(F_k + (1 - \delta)) = 1$$

Consumer (Euler eq) in steady state

$$(1 - \tau_k)\beta(1 + (r - \delta)) = 1$$

and (firm) $r = F_k$. These yield $1 - \tau_k = 1$ so that $\tau_k = 0$. □

A Stronger result

► **Proposition:** Suppose $u(c, l) = \frac{c^{1-\sigma}}{1-\sigma} - v(l)$. Then $\tau_{kt} = 0$ for all $t > 1$.

► **Proof:** In this case $u_{cl} = 0$ and CRRA $u_{cc}c = -\sigma u_c$ so that

$$\begin{aligned} u_c + \mu[u_{cc}c + u_c - u_{cl}l] \\ &= u_c(1 + \mu[\frac{-\sigma u_c}{u_c} + 1]) \\ &= u_c(1 + \mu[1 - \sigma]) \end{aligned}$$

so that the equation above becomes

$$\frac{u_c(t)}{u_c(t+1)} = \beta(F_k(t+1) + (1 - \delta))$$

this + Consumer euler eq immediately implies that $\tau_{kt} = 0$ for all $t > 1$ □

► Chamley (1986) summary:

- The optimal tax: tax capital in period $t = 1$, no capital taxes after that
- Use debt to smooth out tax revenues

Some Difficulties (for now)

- ▶ Period 1 capital tax can be huge
 - ▶ as time period shrinks (goes to continuous time) capital tax in period 1 becomes the same as capital tax in period 0
 - ▶ approaches the level that raises enough revenues to finance all future expenditures, approximates first best
- ▶ Plan to tax capital once and never again not time consistent

Judd (1985)

- ▶ A model of redistribution:
 - ▶ capitalists: own capital stock, do not work, $U(C)$
 - ▶ proletariat: supplies labor inelastically, hands-to-mouth, $u(c)$
- ▶ Government
 - ▶ tax on capital
 - ▶ tax on labor $\Leftrightarrow$ lump-sum to workers
 - ▶ Pareto weight γ on capitalist
- ▶ Two versions:
 - ▶ government can trade debt with capitalists to smooth revenues
 - ▶ government needs to run budget balance

Judd (1985)

► Feasibility

$$c_t + C_t + k_{t+1} = f(k_t) + (1 - \delta)k_t$$

► Workers budget constraint

$$c_t = w_t + T_t = f(k_t) - f'(k_t)k_t + T_t$$

► Capitalist budget constraint

$$C_t + k_{t+1} + b_{t+1} = R_t(k_t + b_t)$$

Capitalists

- Necessary and sufficient conditions (e.g. Ekeland-Scheinkman, 1986)

$$\beta R_t U'(C_t) = U'(C_{t-1})$$

$$\lim_{t \rightarrow \infty} \beta^t U'(C_t)[k_{t+1} + b_{t+1}] = 0$$

- Substitute capitalist's budget constraint

$$\beta U'(C_t)(C_t + k_{t+1} + b_{t+1}) = U'(C_{t-1})(k_t + b_t)$$

$$\lim_{t \rightarrow \infty} \beta^t U'(C_t)[k_{t+1} + b_{t+1}] = 0$$

Planning Problem

- ▶ Planner maximizes weighted sum of utilities

$$\max_{c, C, k, b, T} \sum_{t=0}^{\infty} \beta^t [u(c_t) + \gamma U(C_t)]$$

subject to

$$c_t + C_t + k_{t+1} = f(k_t) + (1 - \delta)k_t$$

$$c_t = f(k_t) - f'(k_t)k_t + T_t$$

$$\beta U'(C_t)(C_t + k_{t+1} + b_{t+1}) = U'(C_{t-1})(k_t + b_t)$$

$$\lim_{t \rightarrow \infty} \beta^t U'(C_t)[k_{t+1} + b_{t+1}] = 0$$

- ▶ Wants redistribution from capitalist to worker
 - ▶ requires sufficiently low γ
 - ▶ for simplicity, will sometimes take the extreme: $\gamma \rightarrow 0$

Judd Version 1: With Public Debt

- Get rid of lump-sum

$$\max_{c, C, k, b} \sum_{t=0}^{\infty} \beta^t [u(c_t) + \gamma U(C_t)]$$

subject to

$$c_t + C_t + k_{t+1} = f(k_t) + (1 - \delta)k_t$$

$$\beta U'(C_t)(C_t + k_{t+1} + b_{t+1}) = U'(C_{t-1})(k_t + b_t)$$

$$\lim_{t \rightarrow \infty} \beta^t U'(C_t)[k_{t+1} + b_{t+1}] = 0$$

- Suppose public debt exists. Substitute away for b_{t+1} and obtain time 0 Implementability condition.
- New planning problem

$$\max_{c, C, k} \sum_{t=0}^{\infty} \beta^t [u(c_t) + \gamma U(C_t)]$$

subject to

$$c_t + C_t + k_{t+1} = f(k_t) + (1 - \delta)k_t$$

$$\sum_{t=0}^{\infty} \beta^t U'(C_t) C_t = U'(C_0) k_0$$

Judd Version 1: With Public Debt

- As before, FOC for C_t :

$$\{\gamma + \mu[\frac{U''(C_t)C_t}{U'(C_t)} + 1]\}U'(C_t) = \lambda_t$$

$$\beta(1 + f'(k_{t+1}) - \delta)\lambda_{t+1} = \lambda_t$$

- So

$$(1 + f'(k_{t+1}) - \delta) = \frac{U'(C_t)}{\beta U'(C_{t+1})} \frac{\gamma + \mu[\frac{U''(C_t)C_t}{U'(C_t)} + 1]}{\gamma + \mu[\frac{U''(C_{t+1})C_{t+1}}{U'(C_{t+1})} + 1]}$$

- In steady state no capital taxes for **any** $\gamma \geq 0$

$$(1 + f'(k_{t+1}) - \delta) = \frac{U'(C^{ss})}{\beta U'(C^{ss})}$$

Judd Version 2: No Public Debt

- ▶ Suppose public debt does not exist
- ▶ Assume U is CES, $U(C) = C^{1-\sigma}/(1-\sigma)$
- ▶ μ_t = multiplier on IC constraint, $\kappa_t = k_t/C_{t-1}$, $v_t = U'(C_t)/u'(c_t)$

$$\mu_0 = 0$$

$$\mu_{t+1} = \mu_t \left(\frac{\sigma - 1}{\sigma \kappa_{t+1}} + 1 \right) + \frac{1}{\beta \sigma \kappa_{t+1} v_t} (1 - \gamma v_t)$$

$$\frac{u'(c_{t+1})}{u'(c_t)} (f'(k_{t+1}) + 1 - \delta) = \frac{1}{\beta} + v_t (\mu_{t+1} - \mu_t)$$

- ▶ Judd (1985) studies interior steady state
 - ▶ for allocation + multipliers
 - ▶ $c_t = c > 0$, $C_t = C > 0$, $k_t = k > 0$, $\mu_t = \mu$
 - ▶ Last FOC $\Rightarrow R^* = 1/\beta$
 - ▶ Capitalists' Euler $\Rightarrow R = 1/\beta$
 - ▶ Hence: **Zero capital tax!**

Straub-Werning (2018)

- ▶ Something is very fishy here
- ▶ Start with Judd V.1, assume CES preferences and $\gamma = 0$
- ▶ Immediately

$$\max_{c, C, k} \sum_{t=0}^{\infty} \beta^t u(c_t)$$

subject to

$$c_t + C_t + k_{t+1} = f(k_t) + (1 - \delta)k_t$$

$$\underbrace{(1 - \sigma) \sum_{t=0}^{\infty} \beta^t U(C_t)}_{\downarrow \text{ in } C \text{ if } \sigma > 1} = \underbrace{U'(C_0)}_{\downarrow \text{ in } C} k_0$$

- ▶ There exists sequence $\{C_t\}$ with $C_t \rightarrow$ that satisfy the last constraint

Judd Version 1 Revisited

- ▶ In the limit $\{C_t\}_t \rightarrow 0$ we are solving the first best

$$\max_{c, C, k} \sum_{t=0}^{\infty} \beta^t u(c_t)$$

subject to

$$c_t + k_{t+1} = f(k_t) + (1 - \delta)k_t$$

- ▶ Hence $\{C_t\}_t \rightarrow 0$ is optimal, the only feasible way is to set tax $\tau_{kt} = 1$ in some t
- ▶ τ_{kt} is a tax on wealth: $\tau_{kt} = 1 \iff$ full expropriation
- ▶ it is equivalent to an infinite tax on interest income.
- ▶ Why Judd v.1 got wrong result? ...Next lecture
- ▶ How about Judd v2 and Chamley (application of Diamond-Mirrlees)?

Bibliography

Chamley, C. (1986). "Optimal Taxation of Capital Income in General Equilibrium with Infinite Lives." *Econometrica*, 54(3), 607–622.

Judd, K. L. (1985). "Redistributive Taxation in a Simple Perfect Foresight Model." *Journal of Public Economics*, 28(1), 59–83.

Ekeland, I., and Scheinkman, J. A. (1986). "Transversality Conditions for Some Infinite Horizon Discrete Time Optimization Problems." *Mathematics of Operations Research*, 11(2), 216–229.

Bibliography (cont.)

Straub, L., and Werning, I. (2018). "Positive Long-Run Capital Taxation: Chamley–Judd Revisited." Working paper (earlier version: NBER Working Paper No. 20441, 2014; published as *American Economic Review*, 110(1), 86–119, 2020).

Goyal, R., Jensen, A., Lagakos, D., and Ndiaye, A. (2025). "Tax Productivity and Economic Development: A Quantitative Macroeconomic Analysis." Working paper